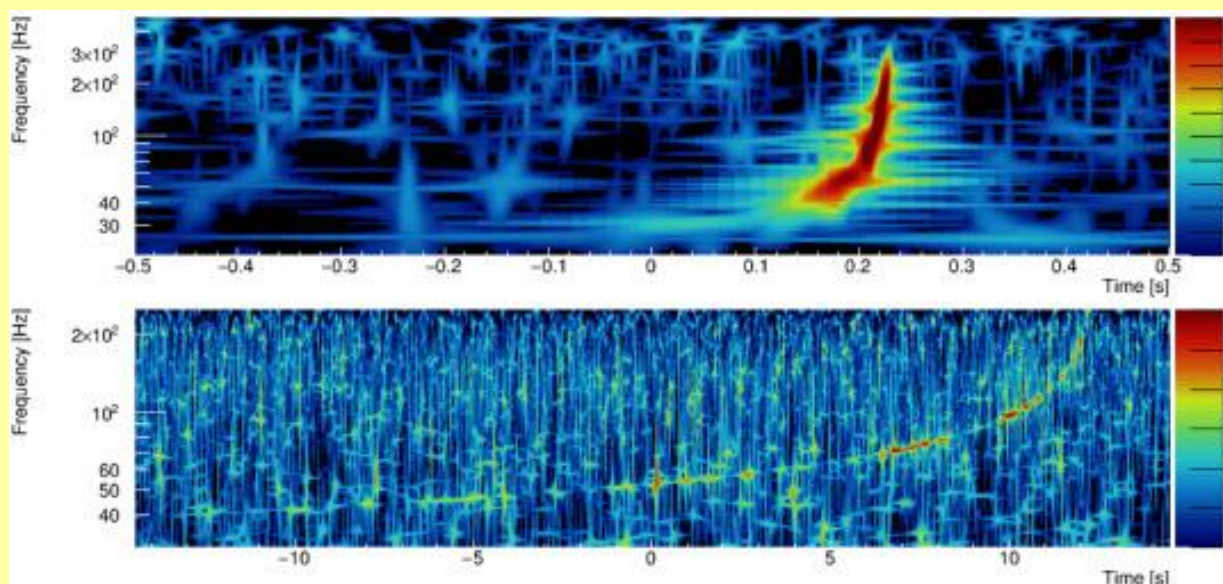


Omicron Pipeline Hyperparameter Tuning using Meta Learning for LIGO O3a run

Meta-Optimized Omniformer

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LIGO's GW data has multiple hyper parameters, somewhere in the range of 10-15 to be tuned to find its statistical inference, Deep learning seems suitable to analyze the effect of these parameters onto the noise classification algorithm based on RF and KarooGP. Finding optimized parameter sets is the main goal of this research project.

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Introduction

Gravitational wave research has found growing interest since its first discovery in 2015 and more so since the team won a Nobel Prize in 2017. With the increasing number of scientist working on the GW data publicly available through GWOSC database, the challenge for working on GW data has shifted from access to processing. Raw gwf files are bulky and even with sufficient sampling take a huge toll on computing resources. To overcome this and work on important data only, OMICRON pipeline was created. This software matches the strain channel data to the auxillary channel's data to find time segments with correlating disturbances, the so called noise transients.

Background:

The main aim of this project is to reduce the manual intervention with the hyperparameter tuning process of OMICRON pipeline and hence optimizing the parameter values to increase noise detection accuracy.

I was inspired by my project in the last semester wherein I had attempted to recreate a study that uses OMICRON to find noise transient in the O1 and O2 run of ligo and classify the noise free and noisy parts of the time series data.

While working on OMICRON myself, I was confused with what values of parameters to set as that would create a threshold for noise transients to be identified. While meeting with the scientists who work on this project, I realized that overall most of the transients are classified by manually setting the hyperparameters with the help of QPlane visualization, something that could be improved.

Necessity:

While OMICRON works excellently, the process is not straightforward. To being with, the raw gwf have to be chosen for any one aux channel for one detector's one run's one sampling rate. Next a CONFIG file for omicron states the thresholds for **frequency range, timing, SNR threshold, mismatch max, Qrange and PSD length**. All of these are additional hyper parameters based on which the output variates.

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Furthermore the output is a hdf5/root/csv file that contains the readings of the AUX channel **for time, frequency ,tstart, tend, fstart, fend, snr, q, amplitude** and phase. All of these are output of the OMICRON pipeline and are provided as per the request CONFIG file.

As can be seen from this description, there are a lot of variable in the equation here and to optimize the output, the best method is to tune the hyper parameters in the best possible manner. In this study we will build one of such methods that employs Deep Learning to maximize noise detection which are characterized by the count of classified noise groups based on their characteristic in the Q plane in main strain.

Idea/Project Description:

The proposed solution is to make use of hyper parameter tuning using Deep Learning methods. Deep Learning methods were specifically chosen as they are faster compared to general purpose AI models and more dynamic compared to ML models. While the classification of the noise still is dependent on the RF and KarooGP models (or any other models of choice), the parameter tuning for Data fetch and Omicron pipeline would be conducted through this proposed solution.

Furthermore, controlling tuning the hyperparameters of the Deep Learning model using an AI model is referred to as META learning and the task of “learning to learn” is satisfied by this approach.

The process of applying this Meta learning approach to the OMICRON analysis is proposed as so:

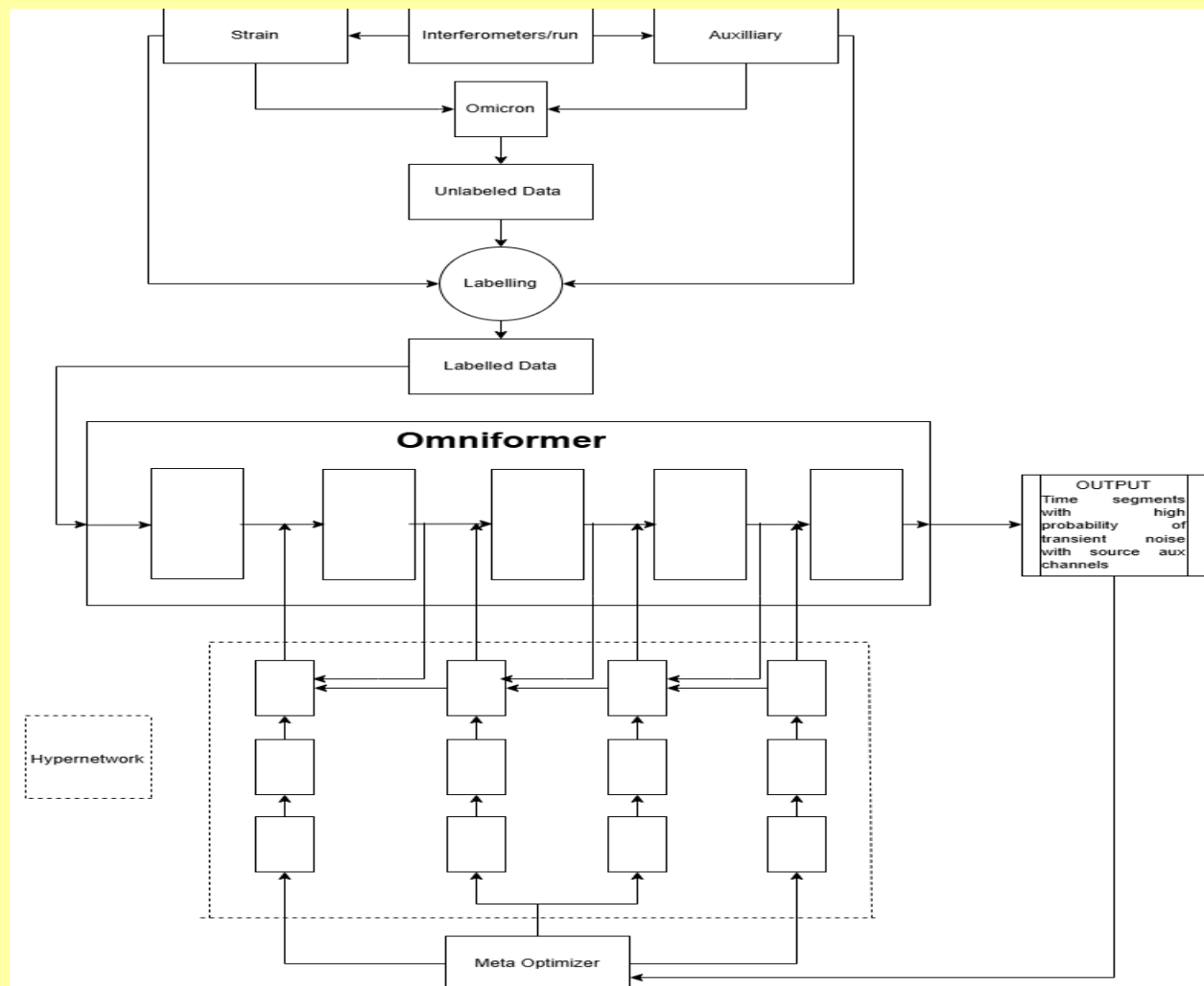
- 1.) OMICRON is run over a diverse set of combinations of parameters to build a dataset.
- 2.) Classification report from the random forest/KarooGP noise classification for the different OMICRON runs is made.
- 3.) These reports are used as training set for the Omniformer model, made from a boiler plate transformer model. A transformer is a general purpose AI model in this case that identifies the trends through a black box for brevity.

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4.) A hypernetwork system tunes the weights of the Transformer based on the outputs of the transformer iteration by iteration.

5.) The weights of the hypernetwork are controlled by the META optimizer using the feedback from last epoch of the Transformer model evaluated over our input data.

This way, a 3-way optimization is applied to the identification of best parameters for omicron. This is better suited to identify transient noise which otherwise could have been easily predicted were it not for their non Gaussian nature.



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Previous Works:

So far, in GW noise identification no use of Meta-learning has been implemented. Various AI models with inherent hyperparameter tuning have been used however none that control the OMICRON parameters as per to my knowledge.

One research to be noted was done on the GW data using Deep filtering that consisted of an analyzer CNN and a predictor CNN both working in sync to predict noise signals in real time. The benefit of this method over the usual matched filtering method in our area of interest was noted with a decrease in about 11% of error for the same SNR levels due to parametric differences between both.

Another benefit of using Meta learning is to overcome non Gaussian nature of these noises which limits the number of statistical models that could be used to characterize them. Some work has been done in using meta models to characterize noise in other fields. Refer the given citations in the end of this document as reference.

Addendum:

<https://doi.org/10.1016/j.neucom.2014.12.100>

<https://arxiv.org/pdf/2408.05151>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9961879>

<https://link.springer.com/content/pdf/10.1007/s10994-018-5710-8.pdf>

<https://arxiv.org/pdf/2408.05151>

<https://doi.org/10.1016/j.neucom.2014.12.100>

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